

Hence, we have

$$\lim_{k \rightarrow +\infty} h(u, v, w) > 0. \quad (16)$$

Therefore,

$$\lim_{k \rightarrow +\infty} \frac{h(u, v, w)}{g(u, v, w)} = \lim_{k \rightarrow +\infty} \frac{\Theta(1)}{\Theta(k)} = 0^+. \quad (17)$$

Therefore,

$$\begin{aligned} \lim_{k \rightarrow +\infty, l \rightarrow 0} \frac{dS_1}{dl} &= \lim_{k \rightarrow +\infty} \iiint_{(u,v,w) \in (c, +\infty)^3: u < v < w} \frac{h(u, v, w)}{g(u, v, w)} dudvdw \\ &= \iiint_{(u,v,w) \in (c, +\infty)^3: u < v < w} \lim_{k \rightarrow +\infty, l \rightarrow 0} \frac{h(u, v, w)}{g(u, v, w)} dudvdw \\ &= \iiint_{(u,v,w) \in (c, +\infty)^3: u < v < w} \lim_{k \rightarrow +\infty, l \rightarrow 0} 0^+ dudvdw \\ &= 0^+. \end{aligned} \quad (18)$$

Hence, for sufficiently large k , we have

$$\lim_{l \rightarrow 0} \frac{d\gamma_{p_{\theta}}^{\mathcal{D}_1/\mathcal{D}}}{dl} = \lim_{l \rightarrow 0} -\frac{dS_1}{dl} < 0. \quad (19)$$

Thus, the proof is completed. $\square$

B Additional Experiment Results

B.1 Additional Experiment of Language Models' *Resistance*

To broadly validate the phenomenon of *Resistance* in language models, we extend the experimental setup described in Section 5.1. Specifically, we use Llama2-7B (Touvron et al., 2023) as the base model and perform a finer-grained snapshot division within *forward alignment*, covering a wider range of k and l . As the evaluation metric, we measure the change in relative KL divergence between the distributions of model θ_l and θ_k , obtained by applying *forward alignment* and *inverse alignment* to slices θ_k and θ_l , respectively, relative to the original distributions of θ_l and θ_k . The experimental results, which is shown in the Table 2, demonstrate that the KL divergence under *inverse alignment* is substantially smaller than that under *forward alignment*, consistent with the conclusions presented in the main text.

Training Steps	θ_1 vs. θ_2	θ_1 vs. θ_3	θ_1 vs. θ_4	θ_1 vs. θ_5	θ_1 vs. θ_6	θ_2 vs. θ_3	θ_2 vs. θ_4
<i>Forward Alignment</i>	0.4568 $\uparrow$	0.4932 $\uparrow$	0.5929 $\uparrow$	0.4439 $\uparrow$	0.4109 $\uparrow$	0.3778 $\uparrow$	0.5580 $\uparrow$
<i>Inverse Alignment</i>	0.2796 $\downarrow$	0.2863 $\downarrow$	0.3960 $\downarrow$	0.3995 $\downarrow$	0.3412 $\downarrow$	0.2752 $\downarrow$	0.3742 $\downarrow$

Table 2: **KL Divergence between different snapshots.** The results show that *forward alignment* exhibits higher values compared to *inverse alignment*, providing evidence for the existence of *Resistance*.

B.2 Additional Experiment of Language Models' *Rebound*

Ablations on the Different Alignment Algorithms. To verify that the consistent *rebound* phenomenon occurs in different alignment algorithms in language models, we conduct language models' *rebound* experiments on reinforcement learning from human feedback (RLHF) (Ouyang et al., 2022), direct preference optimization (DPO) (Rafailov et al., 2024), Kahneman-Tversky optimization (KTO) (Ethayarajh et al., 2024), and simple preference optimization (SimPO) (Meng et al., 2024), using an experimental setup similar to the SFT validation described in Section 5.2. Considering that reinforcement learning algorithms cannot be directly applied to pretrained models, our experimental procedure consists of the following two

steps: (a) SFT of the pretrained model using positive sample data of varying scales as *forward alignment*; (b) *inverse alignment* by applying the corresponding reinforcement learning algorithms on the aligned models, using negative sample data of varying scales, where negative samples serve as selected responses and positive samples serve as rejected responses.

We conduct experiments using the Llama2-7B model on the IMDb task (Maas et al., 2011) with four different RL alignment algorithms. The results, shown in Figure 7, indicate that regardless of the alignment algorithm employed, the model exhibits a decline in performance consistent with that observed in SFT as the amount of positive data increases. Moreover, the rate of this performance degradation accelerates with the increasing volume of positive data. The results further confirm that the broad applicability of language models *rebound* across different alignment algorithms.

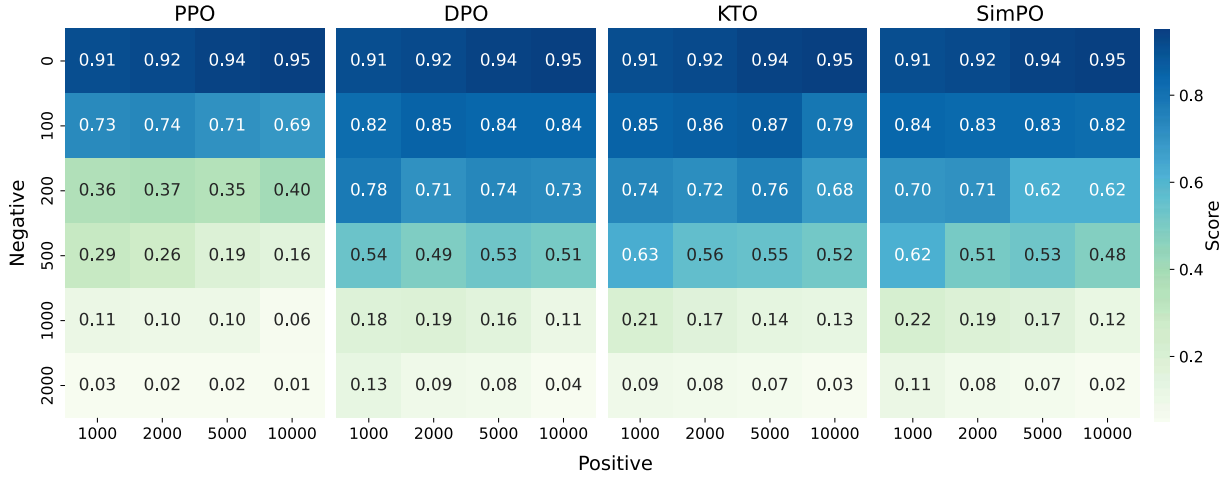


Figure 7: **Experimental results for validating rebound across different alignment algorithms.** The heatmaps from left to right represent the performance of the PPO, DPO, KTO, and SimPO algorithms after fine-tuning with varying amounts of positive and negative data. Darker colors indicate better model performance in the heatmaps. Models trained with more **positive** data perform better initially tend to exhibit worse performance after fine-tuning with **negative** data. The conclusion holds regardless of the alignment algorithm used.

Ablations on the Different Evaluation Metrics. In the main experiments discussed in Section 5.2, we primarily adopt specific evaluation scores to evaluate the *rebound* phenomenon. Using the score models corresponding to alignment targets as the evaluation metrics is the most direct and important method for the task evaluation. However, to further reinforce the robustness of our findings and address the generality of the *rebound* phenomenon, we acknowledge the value of incorporating broader distribution-level metrics beyond task-specific scoring models. In particular, we consider the Kullback-Leibler divergence between aligned models and base models as an additional evaluation metric.

Specifically, we conducted the evaluation through the following procedure: a) We first measure the KL divergence between the aligned model and the base model before and after fine-tuning the base model with safety data. b) We then apply fine-tuning with unsafe data until the KL divergence between the models decreased to a sufficiently small value ϵ (where $\epsilon = 1 \times 10^{-2}$), and record the amount of unsafe data required as an indicator of the difficulty for the model reverting to its pre-trained distribution. The experimental results are reported in Table 3. Under different models and data scales, we observe that only a small amount of unsafe data is needed for a positively fine-tuned model to revert to the pre-training distribution in terms of KL divergence. Furthermore, the larger the amount of safe fine-tuning, the less unsafe data are required. This observation is consistent with the conclusions drawn in Figure 4.

Ablations on the Reverse Finetuning Settings. To rule out the influence of positive data on the *rebound* phenomenon in language models, we conducted a reverse experimental setup: negative data are used during the SFT stage, while positive data are applied during the *inverse alignment* stage. The experimental results, presented in Figure 8, demonstrate that *elasticity* in language models persists under this reverse setting, exhibiting a consistent trend where larger model sizes correspond to greater *elasticity*. The

Base Models	Positive Data Amount			
	1000	2000	5000	10000
Llama2-7B	0.21	0.22	0.26	0.27
Gemma-2B	0.18	0.21	0.24	0.25

(a) KL divergence between models fine-tuned with varying amounts of safety data and the initial models.

Base Models	Positive Data Amount			
	1000	2000	5000	10000
Llama2-7B	961	844	801	729
Gemma-2B	923	853	709	598

(b) Unsafe data amount needed for KL divergence between fine-tuned model and pre-trained model to drop below ϵ .

Table 3: **Experimental results for validating *rebound* on the KL divergence metrics.** The results show that, under different models, the larger the amount of safe fine-tuning, the less unsafe data is required to revert to the pre-training distribution in terms of KL divergence, providing evidence for the existence of *Rebound*.

observation aligns with and further supports the results reported in Figure 5, highlighting the robustness of *rebound* phenomenon across different experimental configurations.

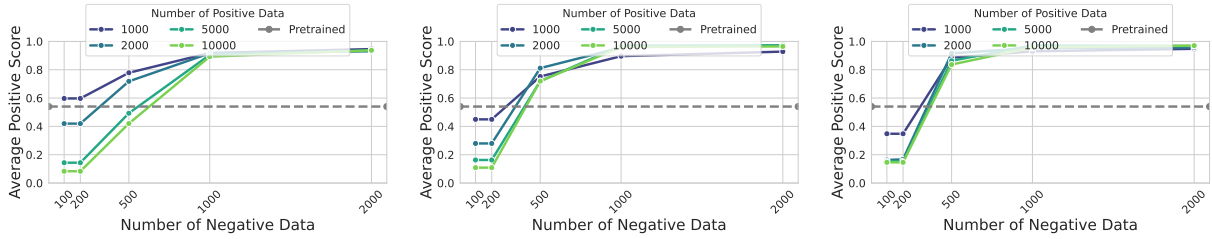


Figure 8: **Reverse Fine-tuning Results on IMDb.** Each subfigure from left to right shows the changes in LLMs with parameter sizes of 0.5B, 4B, and 7B, respectively. Models trained with more **negative** data initially perform worse, but perform better after fine-tuning with **positive** data. As the model size increases, the performance of the aligned model deteriorates more rapidly after fine-tuning with **positive** data.

B.3 Additional Experiment of Internal Factor of Language Models' *Rebound*

Analysis of the Model Size Scale under Different Alignment Algorithms. To verify whether the *rebound* phenomenon in language models becomes more pronounced as model size increases persists across different alignment algorithms, we conduct experiments using the DPO algorithm. Specifically, we employ the Qwen1.5 model series as the base models and follow an experimental setup similar to that described in Section B.2. We carried out experiments on models with parameters 0.5B, 4B, and 7B. The experimental results, presented in Figure 9, demonstrate that *rebound* remains evident under the DPO algorithm, with its strength increasing as model size grows. This finding is consistent with the results reported in Figure 5 and further highlights the generality of *elasticity* in language models.

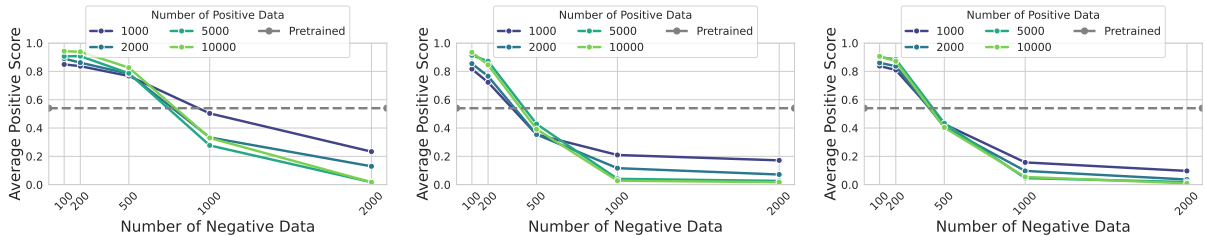


Figure 9: **Experimental results for validating *rebound* increases with model size under DPO fine-tuning.** Each subfigure from left to right shows the changes in LLMs with parameter sizes of 0.5B, 4B, and 7B. As the model size increases, the performance of the aligned model deteriorates more rapidly after fine-tuning with **negative** data.

Analysis of the Pre-training Data Volume. We present the experimental results for a broader range of pre-training data volumes in Figure 10. When the pre-training data volume is 0.1T, 0.5T, and 1.0T, the model still demonstrates the phenomenon that *rebound* increases with the volume of pre-training data, which is consistent with the results reported in Figure 6, where the pre-training data volumes range from